Mohammadreza Ghobadzadeh

Curriculum Vitae

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Research Interests

Control Theory, Photovoltaics and Solar Cell, Power and Energy Systems, Distributed Optimization, Reinforcement Learning, Robotics

Education

2021/24 **M.Sc. in Control Systems Engineering**, GPA: 3.65/4 (17.08/20), University of Tehran, Iran

- Thesis: "*Motion Planning Using Learning-based Model Predictive Control*", under supervision of **Prof. Hamed Kebriaei**, and **Prof. Hadi Moradi.**

2017/21 **B.Sc. in Electrical Engineering**, GPA: 3.92/4 (18.31/20), University of Mohaghegh Ardabili, Ardabil, Iran

- B.Sc. Thesis: "*Stability Analysis of a Mass-Spring System Under Friction Using Lyapunov's Theorem*", under supervision of **Prof. Adel Akbarimajd.**
- Ranked **1**st Among 120 Electrical Engineering B.Sc. Students.

2017/21 **Diploma in Mathematics and Physics**, GPA: 4/4 (19.76/20), Shahid Beheshti High School, Ardabil, Iran

- Affiliated with the National Organization for the Development of Exceptional Talents (**NODET**)

Publications

†Equal contribution

Photovoltaics and Solar Cell Optimization

- **Mohammadreza Ghobadzadeh**†; Hamzeh Kashi Yarandi†; Maryam Shahrostami†; Ali Akbar Haji Zeynali Biyoki; Javad Maleki, "**Beyond Exact Current Matching: Fabrication-Tolerant Design of All-Perovskite Tandem Solar Cells via Surrogate Modeling and Diode-Based J–V Reconstruction,**" Published in **Solar RRL**, **DOI: 10.1002/solr.70434**.

- Hamzeh Kashi Yarandi†; **Mohammadreza Ghobadzadeh**†; Javad Maleki†; Mehrshad Masoudi; Ali Akbar Haji Zeynali Biyoki, "**CHoKI-Based Robust Optimization of All-Perovskite Tandem Solar Cells under Fabrication Variability,**" Energy Conversion and Management: X, revised manuscript under review.

- Javad Maleki†; Hamzeh Kashi Yarandi†; **Mohammadreza Ghobadzadeh**†, "**Risk-Aware Design of Nanostructured All-Perovskite Tandem Solar Cells under Fabrication Variability Using Gaussian Process Surrogates and CVaR Optimization,**" Energy & Environmental Materials, manuscript under review.

Power and Energy Systems

- **Mohammadreza Ghobadzadeh**†; Hamzeh Kashi Yarandi†; Mehran Hajiaghapour-Moghimi, "**Coalition-Based Energy Management of Grid-Connected Smart Homes in the Day-Ahead Electricity Market,**" Scientific Reports, manuscript under review.

- Hamzeh Kashi Yarandi†; **Mohammadreza Ghobadzadeh**†; Mehran Hajiaghapour-Moghimi, "**Uncertainty-Aware Transactive Energy Management of Renewable Interconnected Microgrids with Flexible Cryptocurrency Mining Loads,**" Scientific Reports, manuscript under review.

Control, Optimization, and Intelligent Transportation Systems

- **Mohammadreza Ghobadzadeh**†; Hamzeh Kashi Yarandi†; Abolfazl Yaghmaei; M.J. Yazdanpanah, "**Optimal Control-Based Multi-Agent Lane-Change Reconstruction with Joint IDM–MOBIL Behavioral Inference from NGSIM Trajectories,**" Manuscript prepared for submission to **Transportation Research: Part C**.

- Hamzeh Kashi Yarandi†; **Mohammadreza Ghobadzadeh**†; Abolfazl Yaghmaei; M.J. Yazdanpanah, "**Prediction-Based Barrier-Function Sliding-Mode Control for Nonlinear Systems with Input Delay and Disturbance Estimation,**" Manuscript prepared for submission to **Nonlinear Dynamics**.

Research Experiences

2022-Present **Research Assistant**, University of Tehran, Smart Network Laboratory, under the supervision of **Prof. Hamed Kebriaei**

- **Safe Learning and Learning-Based Model Predictive Control**
  Studying safe control methods for systems with learned policies and uncertain dynamics, inspired by research from Prof. Melanie Zeilinger's group at ETH Zurich. The work focuses on Predictive Safety Filters for reinforcement learning and Gaussian Process-based Model Predictive Control for uncertain systems, with an emphasis on uncertainty propagation, constraint satisfaction, safe and terminal sets, recursive feasibility, and stability.
- **Cooperative Control for Multi-Vessel Ship Towing Using Distributed MPC**
  Conducting research on distributed Model Predictive Control (MPC) for cooperative multi-vessel ship towing systems, with emphasis on coordinated towing-force allocation, trajectory tracking, speed regulation, and collision avoidance through a multi-layer control architecture. Advanced learning-based approaches are investigated to enhance maneuverability in dynamic environments.

2024-Present **Research Assistant,**, University of Tehran, Advanced Control Systems Lab (ACSL), under the supervision of **Prof. Mohammad Javad Yazdanpanah** and **Prof. Abolfazl Yaghmaei**

- **Interaction-Aware Optimal-Control Reconstruction of Lane-Change Trajectories**
  Integrating NGSIM data preprocessing with a multi-agent constrained optimal-control framework for joint reconstruction of the ego vehicle and relevant surrounding vehicles. Incorporating individualized Enhanced Intelligent Driver Model (EIDM) car-following behavior, physical and interaction constraints, and MOBIL-based lane-change incentive and safety information to generate dynamically feasible trajectories while improving physical and interaction consistency.
- **Prediction-Based Barrier-Function Sliding-Mode Control for Nonlinear Systems with Input Delay and Disturbance Estimation**
  Developing a prediction-based sliding-mode control framework for nonlinear systems with input delay and unknown disturbances. Combining nonlinear disturbance observation and future disturbance estimation with a new barrier-function SMC to compensate for delayed actuation and provide robust control without requiring prior knowledge of the disturbance bound in controller implementation. Analyzing sliding-variable invariance, prediction-error bounds, and closed-loop robustness, with extensions toward multichannel and robotic systems.

2025-Present **Research Assistant,**, Shahid Beheshti University, under the supervision of **Prof. Mehran Hajiaghapour-Moghimi**

- **Coalition-Based Energy Management of Grid-Connected Smart Homes in the Day-Ahead Electricity Market**
  Investigating coalition-based energy management for smart homes with renewable generation, battery storage, and flexible loads using a two-stage stochastic MILP. Developing a tunable inter-household energy-exchange mechanism to study the trade-off between cooperation, operating cost, and grid-level performance, with scalability demonstrated for communities of up to 200 homes.
- **Uncertainty-Aware Transactive Energy Management of Renewable Interconnected Microgrids with Flexible Cryptocurrency Mining Loads**
  Developing an uncertainty-aware transactive energy management framework for renewable interconnected microgrids, modeling cryptocurrency mining as a flexible, profit-driven demand resource. Combining stochastic programming and IGDT to coordinate energy storage, local trading, and flexible mining under renewable and electricity-price uncertainty, improving renewable utilization, energy balancing, and system profitability.

Teaching Experience and Workshops

Teaching Assistantships

- **Teaching Assistant**, Distributed Optimization and Learning, Advanced Robotic, Nonlinear Control, Optimal Control, Effective Techniques in Control Systems, [**University of Tehran**]
- **Teaching Assistant**, Model Predictive Control, [**K. N. Toosi University of Technology**]
- **Teaching Assistant**, Linear Control, [**University of Mohaghegh Ardabili**]

Workshops, Educational Materials and Research Mentorship

- **CasADi Workshops**. Designed and conducted advanced workshops on CasADi at the University of Tehran, K. N. Toosi University of Technology, and the University of Mohaghegh Ardabili. The workshops introduced symbolic modeling, numerical optimization, and optimal-control-oriented applications through practical engineering examples.
- Developed a comprehensive CasADi workshop presentation demonstrating the framework, its capabilities, and its applications in control, optimization, and engineering systems. Designed and refined hands-on exercises to help students implement optimization-based models and solve practical problems.
- Designed and refined hands-on educational exercises for MATLAB in the Nonlinear Control course and for CVX, CVXPY, and YALMIP in the Distributed Control course, enabling students to implement numerical optimization, convex optimization, and distributed control methods.
- Mentored students in developing high-quality course projects based on CasADi, MATLAB, CVX, CVXPY, and YALMIP, including technical reports, references, and implementation codes.
- **Publicly instructional videos and implementation codes:** Optimal Control with CasADi, Distributed Optimization and Learning with CasADi & CVXPY, Model Predictive Control with CasADi & YALMIP, Codes.

Selected Courses

Distributed Optimization and Learning, **Advanced Robotics**, **Machine Learning**, **Advanced Instrumentation**, **Estimation and System Identification**, **Nonlinear Control**, **Optimal Control**

Skills

- **Programming Languages**
  Expert In: Python | Matlab
  Familiar with: C++
- **Optimization Tools and Mathematical Libraries**
  Expert In: CasADi | YALMIP | GAMS | CVXPY |CVX
  Familiar with: Pyomo | Do-MPC | Julia
- **Machine Learning and Data Science**
  Expert In: TensorFlow | scikit-learn | GPyTorch | Keras
  Familiar with: OpenAI Gym | PyTorch | GPy
- **Modeling and Simulation Tools**
  Expert In: Simulink | Gazebo
  Familiar with: CARLA | COMSOL Multiphysics
- **Electronic Design and Circuit Simulation**
  Familiar with: Altium Designer | LTspice
- **Robotics and Embedded Systems**
  Expert In: ROS | STM32
  Familiar with: NVIDIA Jetson | Raspberry Pi
- **Documentation Skills**
  LaTeX | Microsoft Office Suite

Languages

- **English**: Fluent
- **Persian**: Native
- **Turkish**: Intermediate

References

Prof. Hamed Kebriaei

- Professor, Department of Electrical and Computer Engineering, University of Tehran
- Email: kebriaei@ut.ac.ir

Prof. Mohammad Javad Yazdanpanah

- Professor, Department of Electrical and Computer Engineering, University of Tehran
- Email: yazdan@ut.ac.ir

Prof. Abolfazl Yaghmaei

- Assistant Professor, Department of Electrical and Computer Engineering, University of Tehran
- Email: yaghmaei@ut.ac.ir

Prof. Mehran Hajiaghapour-Moghimi

- Assistant Professor, Department of Electrical and Computer Engineering, Shahid Beheshti University
- Email: m_hajiaghapour@sbu.ac.ir